Exercise 8G

Understanding

1–3 | 2(½), 3

Hint for Q1: recall that $a^0 = 1$ (except for $a = 0$)

1 State the missing values in this table.

x	0	1	2	3	4	5
2^x						
3^x						243
4^x					256	
5^x		5				
10^x			100			

2 State the value of the unknown number for each statement.

a 2 to the power of what number gives 16?

b 3 to the power of what number gives 81?

c 7 to the power of what number gives 343?

d 10 to the power of what number gives 10 000?

3 Give these numbers as fractions.

a 0.0001 **b** 0.5 **c** 2^{-2} **d** 3^{-3}

Fluency

4–6(½), 8(½) | 4–8(½)

Example 18 Writing equivalent statements involving logarithms

Write an equivalent statement to the following.

a $\log_{10} 1000 = 3$ **b** $2^5 = 32$

Solution

a $10^3 = 1000$

b $\log_2 32 = 5$

Explanation

$\log_a y = x$ is equivalent to $a^x = y$.

$a^x = y$ is equivalent to $\log_a y = x$.

Now you try

Write an equivalent statement to the following.

a $\log_{10} 100 = 2$ **b** $3^4 = 81$

4 Write the following in index form.

a $\log_2 16 = 4$ **b** $\log_{10} 100 = 2$

c $\log_3 27 = 3$ **d** $\log_2 \frac{1}{4} = -2$

e $\log_{10} 0.1 = -1$ **f** $\log_3 \frac{1}{9} = -2$

Hint for Q4: If $\log_a y = x$ then $a^x = y, a > 0, y > 0$.

5 Write the following in logarithmic form.

a $2^3 = 8$ **b** $3^4 = 81$

c $2^5 = 32$ **d** $4^2 = 16$

e $10^{-1} = \frac{1}{10}$ **f** $5^{-3} = \frac{1}{125}$

Hint for Q5: If $a^x = y$ then $\log_a y = x, a > 0, y > 0$.

ISBN 978-1-009-593-861

Example 19 Evaluating logarithms

a Evaluate the following logarithms.

i $\log_2 8$ **ii** $\log_5 625$

b Evaluate the following.

i $\log_3 \frac{1}{9}$ **ii** $\log_{10} 0.001$

c Evaluate, correct to three decimal places, using a calculator.

i $\log_{10} 7$ **ii** $\log_{10} 0.5$

Solution	Explanation
a **i** $\log_2 8 = 3$	Ask the question '2 to what power gives 8?' Note: $2^3 = 8$
ii $\log_5 625 = 4$	$5^4 = 5 \times 5 \times 5 \times 5 = 625$
b **i** $\log_3 \frac{1}{9} = -2$	$3^{-2} = \frac{1}{3^2} = \frac{1}{9}$
ii $\log_{10} 0.001 = -3$	$10^{-3} = \frac{1}{10^3} = \frac{1}{1000} = 0.001$
c **i** $\log_{10} 7 = 0.845$ (to 3 d.p.) **ii** $\log_{10} 0.5 = -0.301$ (to 3 d.p.)	Use the log button on a calculator and use base 10. (Some calculators will give log base 10 by pressing the log button.)

Now you try

a Evaluate the following logarithms.

i $\log_2 16$ **ii** $\log_3 243$

b Evaluate the following.

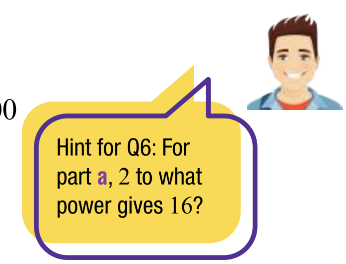
i $\log_2 \frac{1}{8}$ **ii** $\log_{10} 0.01$

c Evaluate, correct to three decimal places, using a calculator.

i $\log_{10} 5$ **ii** $\log_{10} 0.45$

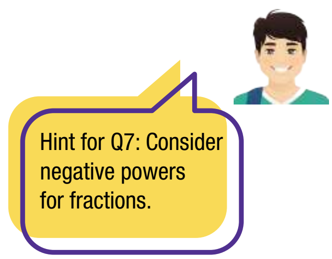
6 Evaluate the following logarithms.

a $\log_2 16$ **b** $\log_2 4$ **c** $\log_2 64$ **d** $\log_3 27$
e $\log_3 3$ **f** $\log_4 16$ **g** $\log_5 125$ **h** $\log_{10} 1000$
i $\log_7 49$ **j** $\log_{11} 121$ **k** $\log_{10} 100\,000$ **l** $\log_9 729$
m $\log_2 1$ **n** $\log_5 1$ **o** $\log_{37} 1$ **p** $\log_1 1$



7 Evaluate the following.

a $\log_2 \frac{1}{8}$ **b** $\log_2 \frac{1}{4}$ **c** $\log_3 \frac{1}{9}$ **d** $\log_{10} \frac{1}{1000}$
e $\log_7 \frac{1}{49}$ **f** $\log_3 \frac{1}{81}$ **g** $\log_5 \frac{1}{625}$ **h** $\log_8 \frac{1}{8}$
i $\log_{10} 0.1$ **j** $\log_{10} 0.001$ **k** $\log_{10} 0.00001$ **l** $\log_2 0.5$
m $\log_2 0.125$ **n** $\log_5 0.2$ **o** $\log_5 0.04$ **p** $\log_3 0.\dot{1}$



ISBN 978-1-009-593-861

8G

8 Evaluate, correct to three decimal places, using a calculator.

a $\log_{10} 6$ b $\log_{10} 47$ c $\log_{10} 162$

d $\log_{10} 0.8$ e $\log_{10} 0.17$ f $\log_{10} \frac{1}{27}$

Problem-solving and reasoning

9,10 | 9–11

9 A single bacterium cell divides into two every minute.

a Complete this cell population table.

Time (minutes)	0	1	2	3	4	5
Population	1	2				

b Write a rule for the population, P, after t minutes.

c Use your rule to find the population after 8 minutes.

d Use trial and error to find the time (correct to the nearest minute) for the population to rise to 10 000.

e Write the exact answer to part **d** as a logarithm.

10 Consider a bacteria population growing such that the total increases 10-fold every hour.

a Complete this table for the population (P) and $\log_{10} P$ for 5 hours (h).

h	0	1	2	3	4	5
P	1	10	100			
$\log_{10}P$						

b Plot a graph of $\log_{10} P$ (y-axis) against hours (x-axis). What do you notice?

c Find a rule linking $\log_{10} P$ with h.

11 Is it possible for a logarithm (of the form $\log_a b$) to give a negative result? If so, give an example and reasons.

The Richter Magnitude

— | 12

12 The Richter magnitude of an earthquake is determined from a logarithm of the amplitude of waves recorded by a seismograph. It uses log base 10. So for example, an earthquake of magnitude 3 is 10 times more powerful than one with magnitude 2 and an earthquake of magnitude 7 is 100 times more powerful than one with magnitude 5.

a Write the missing number. An earthquake of magnitude 6 is:

i ☐ times more powerful than one of magnitude 5.

ii ☐ times more powerful than one of magnitude 4.

iii ☐ times more powerful than one of magnitude 2.

b Write the missing number. An earthquake of magnitude 9 is:

i ☐ times more powerful than one of magnitude 8.

ii 1000 times more powerful than one of magnitude ☐.

iii 10^6 times more powerful than one of magnitude ☐.

ISBN 978-1-009-593-861